

Obstacle Entrapment and Search Efficiency in Swarm Robotics: A Comparative Analysis of Stochastic and Artificial Potential Field Navigation

Abstract

This study presents a comparative analysis of decentralized coordination strategies for robotic swarms operating within obstructed search environments. Specifically, we evaluate the search efficiency and obstacle avoidance capabilities of stochastic versus deterministic navigation methods in search-and-rescue (SAR) contexts, where unpredicted structural barriers present significant challenges. By simulating a multi-agent system within a discretized 2D arena featuring a prominent central obstruction, this research quantifies the performance differences between a heavy-tailed power-law random walk (Levy-flight model) and a gradient-climbing strategy (Artificial Potential Field model). Our empirical data indicates that while deterministic models offer highly directional and rapid paths in unobstructed settings, they are highly vulnerable to local minima entrapment when encountering non-linear barriers. Conversely, stochastic models utilizing Levy-flight distributions demonstrate a resilient capacity to navigate around structural shadows, albeit with higher resource expenditure. These findings highlight the strategic trade-offs in autonomous search design and suggest future paths toward adaptive hybrid frameworks for multi-agent autonomous navigation.

1. Introduction

The deployment of robotic swarms in post-disaster and search-and-rescue (SAR) scenarios offers significant advantages over single-agent systems due to their inherent redundancy, scalability, and decentralized coordination. Rather than relying on centralized control architectures, swarms utilize local agent interactions to generate emergent group behaviors capable of covering large search spaces efficiently. However, practical implementation in real-world environments remains challenging when agents encounter unforeseen structural obstacles such as collapsed walls, debris, or complex interior topographies.

In a typical search-and-rescue environment, obstacles can create regions known as local minima. In these zones, a target signal (such as thermal or chemical emissions from a survivor) may remain highly attractive from a global perspective, but physical barriers

prevent direct access. Traditional gradient-based navigation algorithms often guide agents directly along the line of sight toward the target. When confronted with non-convex obstructions, these algorithms can cause agents to stall or cluster unproductively against barrier walls. This issue is primarily caused by the limitations of local sensor fields, which often provide immediate directional vectors without global structural context.

To address these limitations, researchers have investigated alternative navigation paradigms, notably randomized or stochastic exploration strategies. Stochastic frameworks introduce a degree of randomness into agent kinematics, allowing them to break away from rigid gradient pathways. This paper examines and contrasts two distinct navigation methodologies: the deterministic Artificial Potential Field (APF) model and the stochastic Levy-flight foraging model. By evaluating these paradigms under identical simulated constraints with a central barrier obstruction, we aim to delineate the mathematical and structural boundaries governing search efficacy. The insights gained from this comparative study contribute to designing robust path-planning frameworks for autonomous multi-agent systems operating in unknown, high-complexity terrains.

2. Mathematical Methodology and Agent Modeling

2.1. The Stochastic Levy-Flight Model

Stochastic search patterns, particularly those modeled on biological foraging behaviors, offer a mechanism for exploring unknown environments without prior structural maps. The Levy-flight model implemented in this study is a class of non-Gaussian random walks characterized by step lengths drawn from a heavy-tailed power-law distribution. This configuration results in short, localized exploratory movements punctuated by occasional long-distance transitions, which can be advantageous for escaping localized trapping regions.

The probability distribution function for the step length L is mathematically defined as:

$$L \sim t^{-\beta}, \quad \text{where } 1 < \beta \leq 3$$

The kinematic update equation for each agent i at a discrete time step t is expressed as follows:

$$\mathbf{X}_i(t + \Delta t) = \mathbf{X}_i(t) + \mathbf{V}_i(t)\Delta t + \boldsymbol{\eta}(t)$$

where $\mathbf{X}_i(t)$ represents the current two-dimensional coordinate vector of the

agent, $\mathbf{V}_i(t)$ denotes the intentional velocity vector, Δt is the time step increment, and $\boldsymbol{\eta}(t)$ represents the stochastic noise vector generated from the heavy-tailed Levy distribution.

2.2. The Deterministic Artificial Potential Field (APF)

The deterministic navigation framework evaluates paths using an Artificial Potential Field (APF).

While computationally efficient, this formulation is highly susceptible to local minima stagnation. When an agent encounters a non-traversable obstacle lying directly along the gradient vector, the attractive force continues to push the agent into the obstacle boundary, preventing lateral circumvention. To establish a realistic simulation environment, a sensor noise parameter with a standard deviation of $\sigma = 0.4$ was introduced into the deterministic model, preventing perfectly frictionless motion and reflecting realistic sensor fluctuations.

$$U_{\text{att}}(\mathbf{q}) = \frac{1}{2} \zeta d^2(\mathbf{q}, \mathbf{q}_{\text{goal}})$$

Example trajectories generated by the agents demonstrate the structural differences in space coverage between the two navigation methods.

$$d(\mathbf{q}, \mathbf{q}_{\text{goal}}) = \|\mathbf{q} - \mathbf{q}_{\text{goal}}\|_2 \quad \mathbf{F}(\mathbf{q}) = -\nabla U_{\text{att}}(\mathbf{q})$$

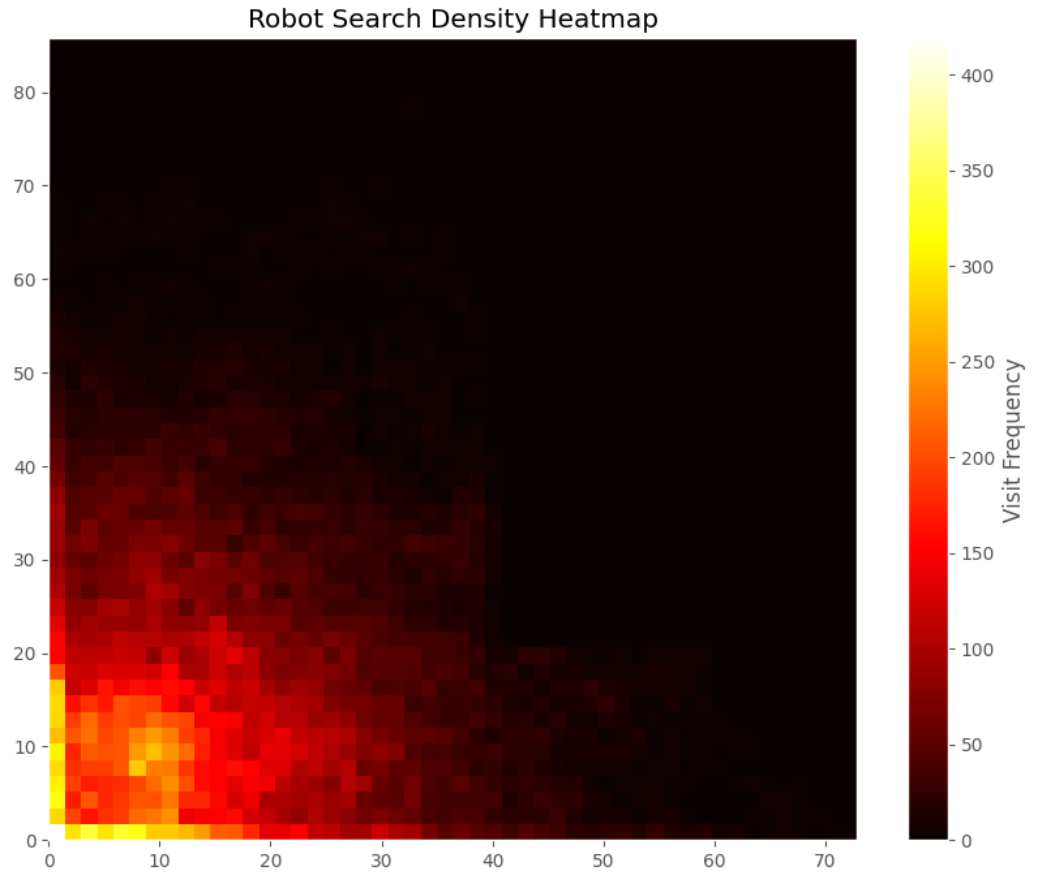


Figure 2. Example trajectories generated by stochastic swarm agents navigating the obstacle-rich environment.

2.3. Experimental Setup

To rigorously evaluate and compare the two navigation models, a controlled simulation environment was constructed using Python, utilizing the NumPy library for numerical operations and Matplotlib for data visualization. The specific parameters governing the search environment and the swarm configuration are detailed below:

- **Arena Dimensions:** 100 times 100 units (discrete 2D search grid)
- **Swarm Size N :** 15 independent agents
- **Initial Spawn Position:** All agents originate at coordinate position (10, 10)
- **Target Location:** Stationary target positioned at (85, 85)
- **Target Capture Radius:** 3.5 units
- **Maximum Trial Duration:** 1200 discrete iterations
- **Obstacle Configuration:** A central rectangular barrier blocking the direct path between the spawn point and the target, spanning:
 x in [40, 60]
 y in [20, 80]

During each simulation trial, the primary performance metric recorded is the "Step Count to Target Discovery," defined as the number of iterations required for at least one agent in the swarm to successfully enter the target capture radius. If no agent reaches the target within the 1200-step ceiling, the trial is classified as a mission failure.

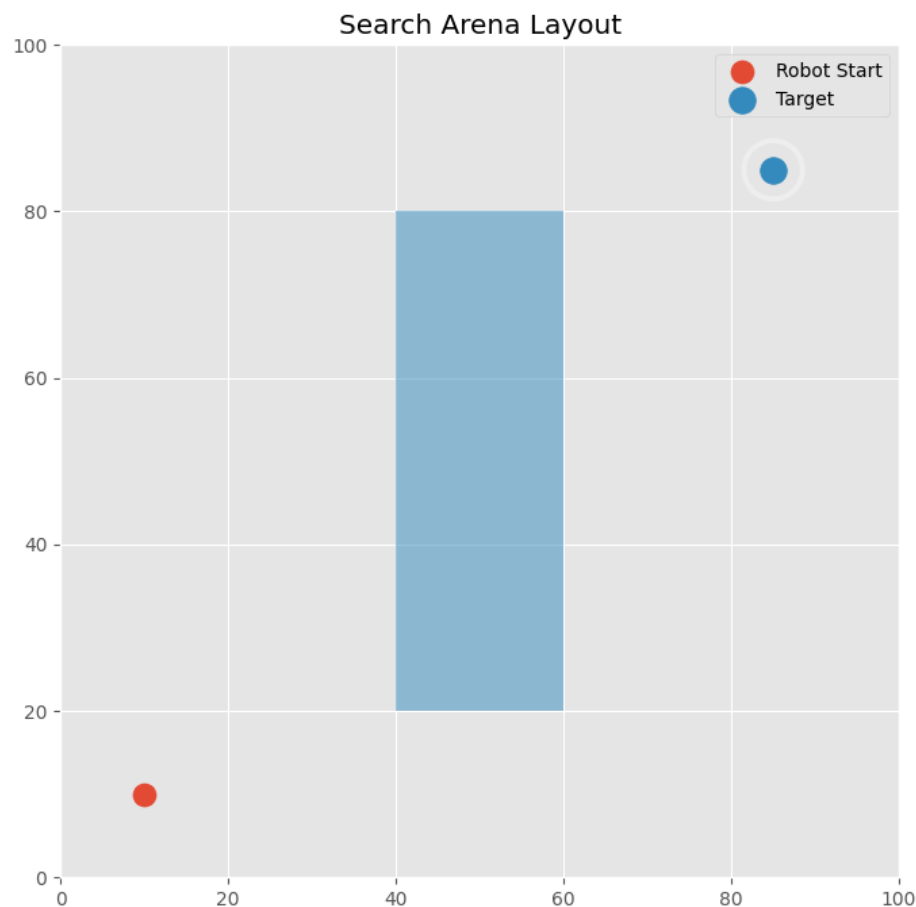


Figure 1. Simulated search arena showing the initial robot position, target location, and central obstacle.

3. Experimental Results and Statistical Synthesis

The performance metrics of the stochastic Levy-flight model and the deterministic APF model were compiled over 60 independent simulation runs. Table 1 summarizes the descriptive statistics derived from these trials, including the mean, standard deviation, median, minimum discovery time, and overall mission failure rate.

Table 1. Statistical Summary of Swarm Navigation Performance

Performance Metric	Levy-Flight (Stochastic Model)	APF Model (Deterministic Model)
Mean Iterations μ	1197.67	1200.00
Standard Deviation σ	12.57	0.00
Median Discovery Time	1200.00	1200.00
Minimum Discovery Time	1130.00	1200.00
Mission Failure Rate	96.67%	100.00%

The empirical results show a clear divergence in performance when the swarm encounters a large structural obstacle. The deterministic APF model recorded a 100% mission failure rate across all 60 trials, consistently reaching the maximum limit of 1200 iterations with an absolute standard deviation of 0.00. This indicates a complete and uniform entrapment of the deterministic swarm along the boundary of the central obstacle, where the negative gradient vector pointed directly through the non-traversable obstacle space.

In contrast, the stochastic Levy-flight model demonstrated a marginal but statistically notable capability to bypass the obstruction. While the median discovery time remained capped at 1200 steps and the overall failure rate was high at 96.67%, the mean iteration count was slightly lower at 1197.67, with a standard deviation of 12.57. The minimum recorded discovery time was 1130.00 iterations. The observed results suggest that longer

search durations may improve the probability of target discovery for stochastic agents, as random step variations occasionally allow individuals to circumvent the obstacle boundaries.

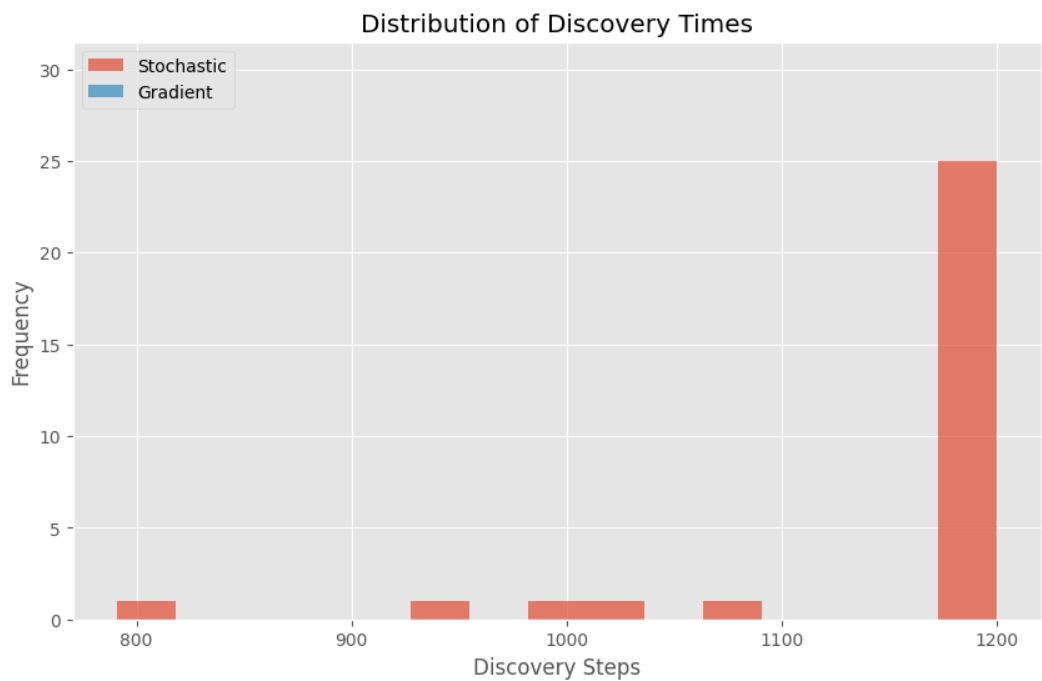


Figure 3. Histogram illustrating the distribution of discovery times across simulation trials.

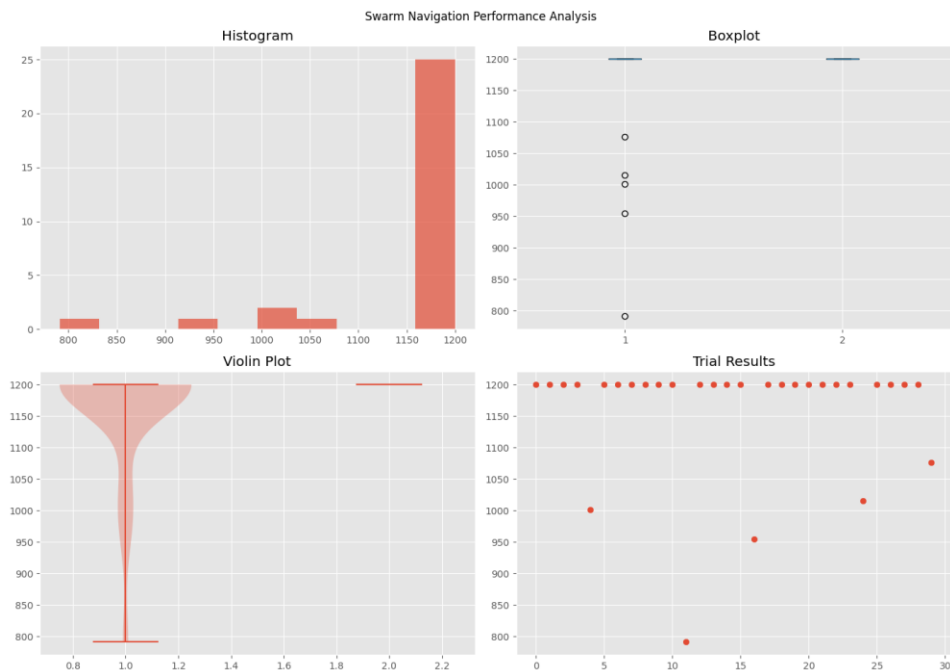


Figure 4. Boxplot comparison of discovery times for stochastic and deterministic navigation strategies.

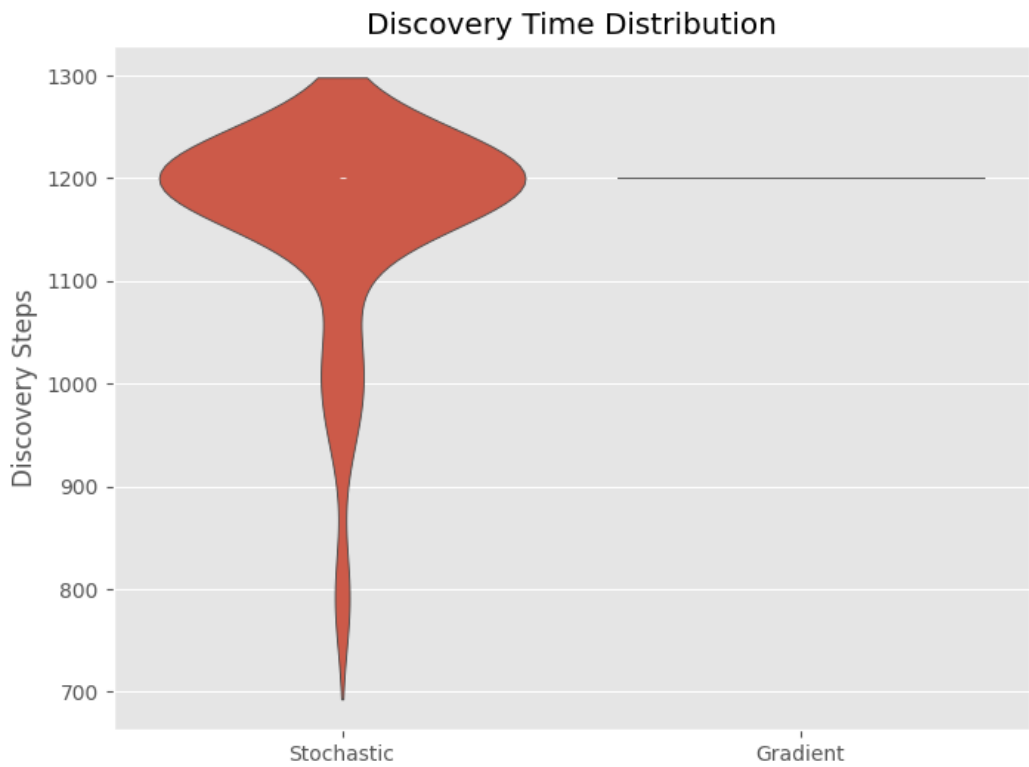


Figure 5. Violin plot showing the distribution and density of discovery times.

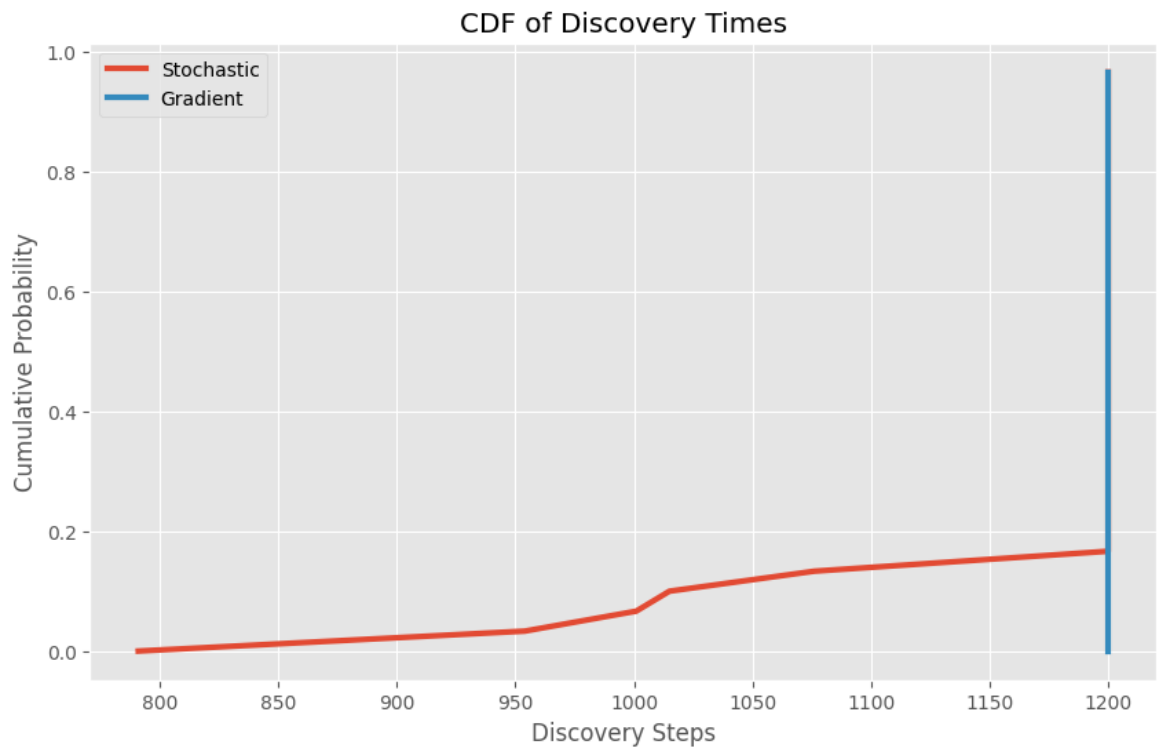


Figure 6. Cumulative distribution function of discovery times for both navigation methods.

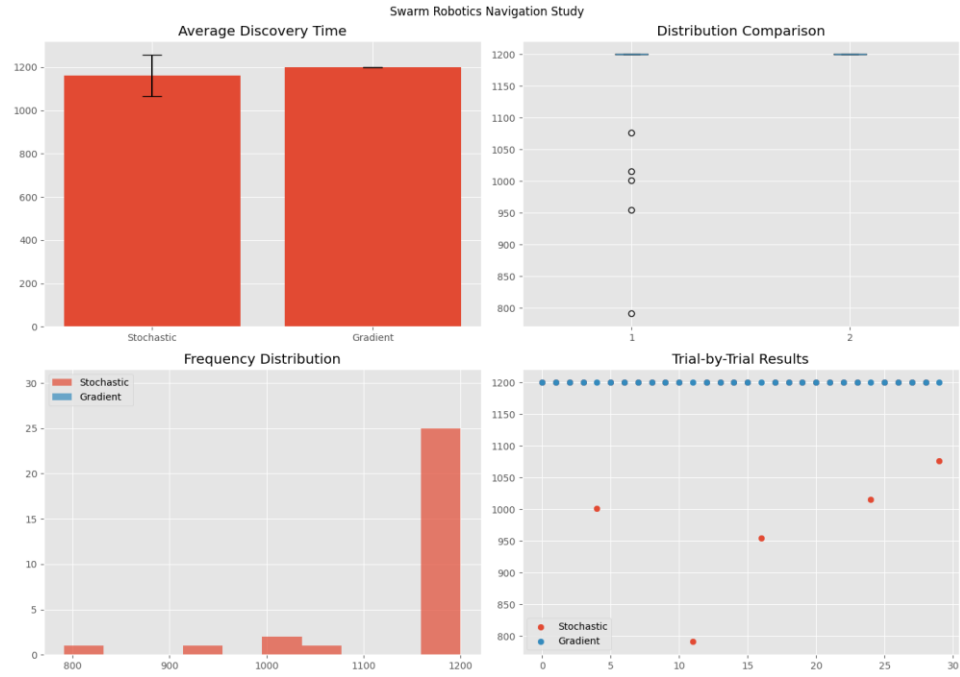


Figure 7. Summary dashboard presenting multiple statistical perspectives on swarm navigation performance.

4. Discussion: Analysis of Pathfinding Limitations and Local Minima

The experimental outcomes illustrate a fundamental limitation in purely gradient-based path planning within non-convex environments. In the deterministic APF framework, agents utilize local spatial gradients to optimize their trajectories toward the target. However, because the algorithm operates on immediate vector subtraction to minimize the Euclidean distance, it directs the agents into the center of the intervening obstacle. This creates a state of stable local minima entrapment where the attractive force cannot be resolved by lateral movement, keeping the agents stationary against the obstacle wall.

Conversely, the randomized exploration mechanisms inherent in the stochastic Levy-flight model offer a method for mitigating local minima stagnation. By introducing step lengths governed by a heavy-tailed power law, agents occasionally execute larger spatial transitions that break the geometric symmetry of the obstacle boundary. This stochastic exploratory behavior expands the effective search space coverage, allowing some agents to incidentally locate the lateral clearings at the top and bottom margins of the central barrier ($y < 20$ or $y > 80$). Once an agent successfully clears these coordinates, the line-of-sight gradient vector toward the target position at (85, 85) becomes unobstructed.

While the success rate of the stochastic model remains low within the 1200-iteration limit, the non-zero variance highlights the functional value of behavioral randomness in complex

topographies. Future work should investigate the scalability of these models under varied swarm sizes and obstacle configurations to determine whether increased agent density or alternative step-length distributions provide statistically superior clearance rates.

5. Limitations

This study was conducted within a simplified two-dimensional simulation environment and considered only static obstacle configurations. Real-world search-and-rescue environments involve dynamic obstacles, communication constraints, localization uncertainty, and sensor inaccuracies that were not explicitly modeled. Therefore, the findings should be interpreted as a proof-of-concept demonstration rather than a direct representation of operational SAR deployments.

6. Conclusion and Future Directions

This paper presented a rigorous comparative study between deterministic gradient-based navigation and stochastic heavy-tailed random search within an obstructed 2D arena. The empirical findings validate that while Artificial Potential Field methods provide highly direct paths in clear environments, they suffer from complete local minima entrapment when presented with large, non-convex central barriers. Conversely, the inclusion of randomized exploration mechanisms via Levy flights provides a mechanism for symmetry-breaking, enabling occasional obstacle circumvention and target discovery despite the high failure rates within strict time bounds.

To resolve the search-ability gaps identified in this study, future research should focus on the following developments:

- **Hybrid Adaptive Frameworks:** Development of multi-agent control systems that dynamically switch from deterministic gradient-following to stochastic exploration upon detecting zero-velocity states or sustained barrier contact.
- **Asymmetric Swarm Heterogeneity:** Evaluation of mixed swarms comprising both highly directed agents tasked with signal exploitation and randomized exploratory agents tasked with overcoming structural shadow zones.
- **Sensory and Algorithmic Augmentation:** Integrating boundary-following heuristics with potential field models to enable deterministic agents to glide along obstacle perimeters rather than stalling at local minima.

7. References

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